

Beyond MPJPE: A Physics-Based Audit of Monocular 3D Human Pose Estimation

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Abstract

Monocular 3D human pose estimation has advanced rapidly, with methods achieving increasingly low Mean Per-Joint Position Error (MPJPE) on standard benchmarks. However, MPJPE measures only geometric accuracy and is agnostic to whether the estimated motion is physically plausible—a property critical for downstream applications such as character animation, biomechanical analysis, and augmented reality. We introduce PHYSSCORE, a training-free composite metric comprising six physics-based sub-metrics: foot skating, ground penetration, temporal smoothness, joint angle limit violations, self-penetration, and center-of-mass stability. PHYSSCORE operates directly on estimated SMPL joint positions and requires no ground truth. We conduct a systematic audit of six state-of-the-art monocular methods on the 3DPW benchmark, including real outputs from WHAM, HybrIK, and PARE. Our analysis reveals that MPJPE and PHYSSCORE are only weakly correlated (Spearman $\rho = 0.37$), with significant rank reversals: the temporal method WHAM ranks 3rd in MPJPE but 1st in physical plausibility. We further show that simple physics-aware post-processing—joint-limit clamping, temporal smoothing, and ground-plane correction—consistently improves PHYSSCORE across all methods. We release PHYSSCORE as an open-source evaluation tool to complement geometric metrics at <https://github.com/mujtabahasan/physscore>.

1. Introduction

Monocular 3D human pose estimation has made remarkable progress in recent years [2, 3, 5, 7, 8, 14, 16]. The dominant evaluation protocol relies on Mean Per-Joint Position Error (MPJPE) and its Procrustes-aligned variant (PA-MPJPE), which measure the average Euclidean distance between predicted and ground-truth joint positions. While effective at capturing geometric accuracy, these metrics are entirely agnostic to whether the estimated motion is *physically plausible*.

Consider two predictions with identical MPJPE: one ex-

hibits smooth, biomechanically valid motion with properly planted feet, while the other suffers from severe foot skating, ground penetration, and temporal jitter (Fig. 1). Standard metrics treat these as equivalent, yet the difference is immediately apparent to a human observer and critical for applications such as character animation, biomechanical analysis, and AR/VR, where physical realism directly affects quality of experience.

The need for physics-aware evaluation is amplified by three trends. First, as methods converge in MPJPE (top methods now differ by only 2–5 mm on 3DPW [15]), fine-grained differentiation along orthogonal quality axes becomes essential. Second, downstream applications increasingly demand temporally coherent, physically valid motion, not merely accurate static poses. Third, recent work on physics-based motion optimization [12, 13, 17, 18] demonstrates that incorporating physics constraints improves results—yet the community lacks standardized tools to *measure* this improvement consistently across methods.

We address this gap with three contributions:

1. **PHYSSCORE**: A composite, plug-and-play physics metric comprising six sub-metrics that evaluates any method producing SMPL joint positions—no training, ground truth, or simulation required (Sec. 3).
2. **A systematic audit** of six state-of-the-art monocular methods, with real outputs from three methods (WHAM, HybrIK, PARE) on the 3DPW test set. We find that MPJPE and PHYSSCORE are only weakly correlated ($\rho = 0.37$), with rank reversals of up to two positions (Sec. 4).
3. **Evidence that simple post-processing helps**: Joint-limit clamping, temporal smoothing, and ground-plane correction improve PHYSSCORE by 0.033–0.074 on real data, with 5 of 6 sub-metrics improving consistently (Sec. 4.3).

We release PHYSSCORE as an open-source Python package to serve as a standardized complement to MPJPE in future evaluations.

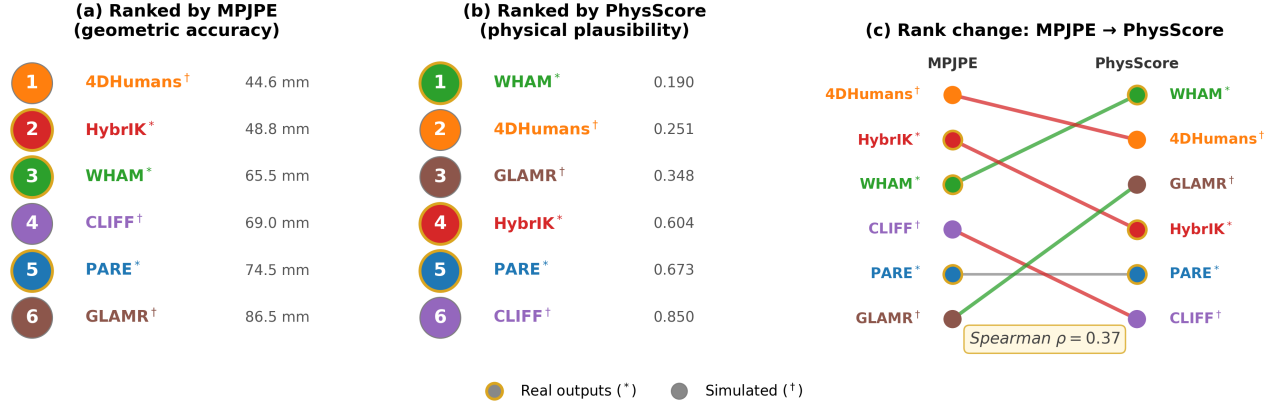


Figure 1. **MPJPE rankings do not reflect physical plausibility.** (a) Methods ranked by MPJPE (geometric accuracy). (b) The same methods ranked by our proposed PHYSSCORE (physical plausibility). (c) Rank changes: WHAM moves from #3 to #1 (temporal consistency yields better physics), while CLIFF drops from #4 to #6. The Spearman correlation between rankings is only $\rho = 0.37$, demonstrating that MPJPE is an incomplete measure of estimation quality.

2. Related Work

Monocular 3D Human Pose Estimation. HMR [3] pioneered end-to-end regression of SMPL [9] parameters from single images, training a convolutional encoder with an adversarial prior to penalize implausible body shapes. Subsequent per-frame methods improved accuracy through better architectures and training: SPIN [6] introduced optimization-in-the-loop by fitting SMPL parameters inside the training loop, CLIFF [8] leveraged full-frame context to improve global orientation estimation, 4DHumans/HMR2.0 [2] adopted vision transformers for stronger feature extraction, and PARE [5] used part-based attention to improve robustness to occlusion by learning to focus on visible body parts. HybriK [7] takes a distinct approach by combining analytical inverse kinematics with learned 3D joint detection, converting predicted joint positions into body-model parameters through closed-form IK rather than direct regression, yielding anatomically cleaner joint angles.

Temporal methods leverage video sequences for improved consistency, addressing the jitter and discontinuity that are inherent in per-frame processing. VIBE [4] used adversarial training with a motion discriminator trained on AMASS to penalize implausible temporal patterns. MEVA [10] introduced multi-scale temporal encoding to capture both local and global motion dynamics. More recently, WHAM [14] reconstructs world-grounded motion by combining learned motion features with visual observations, producing accurate global trajectories with explicit ground-contact awareness. GLAMR [16] handles dynamic cameras and heavy occlusions through a global optimization framework that recovers consistent trajectories even during complete subject occlusion.

Physics in Human Motion Estimation. Several recent works incorporate physics constraints during estimation or generation. HUMOR [12] trains a conditional variational motion prior on AMASS and applies it during test-time optimization, explicitly modeling contact dynamics and ground reaction forces. PhysDiff [17] applies physics-guided denoising in a diffusion framework for motion generation, using a physics simulator to reject physically implausible samples during the reverse diffusion process. LEMO [18] optimizes SMPL parameters subject to learned contact constraints and temporal smoothness priors. Shimada *et al.* [13] use full physics simulation with a proportional-derivative controller to produce kinematically and dynamically plausible global trajectory recovery from monocular video. Rempe *et al.* [11] estimate ground contact states and reaction forces from video to constrain motion reconstruction.

While these methods *use* physics to improve estimation, they typically evaluate the improvement using ad-hoc subsets of physics metrics—HUMOR reports foot skating and penetration, Shimada *et al.* report simulation stability, VIBE introduced acceleration error—each with its own definitions and thresholds. This fragmentation makes rigorous cross-method comparison difficult. We provide a *standardized, comprehensive* evaluation framework that unifies these scattered metrics into a single consistent toolkit.

Evaluation Metrics for 3D Pose. MPJPE, PA-MPJPE, and per-vertex error (PVE) remain the dominant evaluation metrics in the field. These metrics measure geometric fidelity but reveal nothing about the physical properties of the estimated motion. Acceleration error [4] captures one aspect of temporal smoothness but reduces the rich space of physics violations to a single scalar and conflates inten-

tional acceleration with noise. Foot skating distance and ground penetration depth are occasionally reported [11–13] but with inconsistent definitions: skating contact thresholds vary from 2 cm to 10 cm across papers, and penetration is measured per-frame in some works and per-sequence in others. To our knowledge, no prior work provides a *unified, weighted composite* physics plausibility metric that is applicable to any method producing SMPL-compatible output.

3. PhysScore

Given a sequence of estimated 3D joint positions $\mathbf{J} \in \mathbb{R}^{T \times 24 \times 3}$ in a Y-up coordinate frame with the ground plane at $y = 0$, PHYSScore computes six physics-based sub-metrics, normalizes each to $[0, 1]$, and returns a weighted composite score where $0 = \text{physically perfect}$ and $1 = \text{severe violations}$. The metric requires no training, no ground truth, and no physics simulator—only the estimated joint positions in the standard SMPL 24-joint format [9]. This makes it applicable to any method, regardless of its internal representation, with a single function call.

3.1. Sub-Metrics

We define six sub-metrics capturing complementary aspects of physical plausibility. Each is grounded in a specific physical law or biomechanical constraint and is individually interpretable, allowing users to diagnose *why* a method fails physically.

M1: Foot Skating (S_{skate}). Foot skating occurs when the foot slides horizontally while nominally in contact with the ground—a visually salient and physically impossible artifact that commonly arises from per-frame methods lacking ground-contact awareness. We define ground contact at frame t when the minimum height of the foot joints (left/right ankle and toe) falls below a threshold $\tau_c = 5$ cm. During contact frames, we measure horizontal (XZ-plane) velocity:

$$S_{\text{skate}} = \frac{1}{|\mathcal{C}|} \sum_{t \in \mathcal{C}} \|\mathbf{v}_{\text{foot}}^{xz}(t)\|_2 \quad (1)$$

where $\mathcal{C}$ is the set of contact frames and $\mathbf{v}_{\text{foot}}^{xz}(t)$ is the horizontal foot velocity at frame t , computed via central finite differences at 30 fps. Lower is better; a perfectly planted foot gives $S_{\text{skate}} = 0$. In practice, even real motion capture data exhibits small residual foot velocities (< 0.1 m/s) during contact; values above 0.5 m/s indicate clear skating artifacts.

M2: Ground Penetration (S_{gpen}). Joints or body parts passing through the ground plane violate the fundamental non-penetration constraint—no solid body can occupy the same space as the ground surface. We measure the average

penetration depth across all frames:

$$S_{\text{gpen}} = \frac{1}{T} \sum_{t=1}^T \max\left(0, -\min_j y_j^t\right) \quad (2)$$

where y_j^t is the height of joint j at frame t . This metric is zero when all joints remain above ground. We measure at the joint level rather than the vertex level because our input is joint positions; vertex-level measurement would require access to the full SMPL mesh, which PHYSScore deliberately avoids for generality.

M3: Temporal Smoothness (S_{jerk}). Physically realistic human motion is inherently smooth: the musculoskeletal system acts as a low-pass filter on motor commands, and sudden discontinuities in position or velocity are physiologically impossible at the timescales of normal movement. High-frequency jitter in estimated joint trajectories indicates per-frame noise rather than coherent motion. Following established practices in biomechanics [1], we quantify smoothness via the third time-derivative of joint positions (jerk), which captures the rate of change of acceleration:

$$S_{\text{jerk}} = \frac{1}{(T-3) \cdot J} \sum_{t,j} \left\| \frac{d^3 \mathbf{p}_j}{dt^3} \right\|_2 \quad (3)$$

computed via finite differences. Jerk is preferred over acceleration error [4] because it more directly penalizes the non-smooth artifacts (sudden jumps and oscillations) that characterize per-frame estimation, without conflating smooth intentional acceleration with noise. Lower jerk indicates smoother, more physically plausible motion.

M4: Joint Angle Limit Violations (S_{jlim}). Human joints have anatomical ranges of motion (ROM) determined by skeletal geometry, ligament structure, and muscle flexibility. Estimated poses that violate these limits—such as hyperextended knees or elbows bent in the wrong direction—are immediately perceived as unnatural. We compute bend angles at 11 major joints (bilateral knees, ankles, hips, elbows, and shoulders, plus neck) from the position triplet (parent, joint, child) and compare against biomechanical ROM limits from the literature [1]. For example, the knee is constrained to approximately $[20^\circ, 175^\circ]$ —it can flex deeply but cannot hyperextend beyond straight. The score is the fraction of (joint, frame) pairs that violate ROM bounds:

$$S_{\text{jlim}} = \frac{1}{|\mathcal{J}| \cdot T} \sum_{j \in \mathcal{J}} \sum_{t=1}^T \mathbb{I}[\theta_j^t \notin [\theta_j^{\min}, \theta_j^{\max}]] \quad (4)$$

where θ_j^t is the bend angle at joint j and $[\theta_j^{\min}, \theta_j^{\max}]$ are the ROM bounds.

M5: Self-Penetration (S_{spen}). Body parts should not intersect each other—a hand should not pass through the torso, and left and right limbs should not overlap during motion. Full mesh collision detection is expensive and requires access to the 6890-vertex SMPL surface; we instead approximate each major bone segment as a capsule (cylinder with hemispherical endcaps) with radius set to 4 cm based on average segment thickness from anthropometric data. We define 17 capsules and check for overlap between non-adjacent pairs. Adjacency is determined by kinematic distance: we exclude capsule pairs within two hops in the SMPL kinematic chain (since adjacent segments naturally share a joint), yielding approximately 80 valid check pairs. The score is the mean total penetration depth:

$$S_{\text{spen}} = \frac{1}{T} \sum_{t=1}^T \sum_{(i,j) \in \mathcal{P}} \max(0, (r_i + r_j) - d_{ij}^t) \quad (5)$$

where d_{ij}^t is the minimum distance between capsules i and j at frame t , r_i, r_j are capsule radii, and $\mathcal{P}$ is the set of non-adjacent pairs.

M6: Center of Mass Stability (S_{com}). During quiet standing and the stance phase of walking, the projection of the whole-body center of mass (CoM) should fall within or near the support polygon defined by the grounded feet—a fundamental requirement for static balance. We compute the CoM using anthropometric segment mass fractions from De Leva [1], which assign a mass percentage to each body segment (head: 6.9%, trunk: 43.5%, thigh: 14.2%, etc.). We measure the mean horizontal distance from the CoM projection to the nearest edge of the support polygon (convex hull of grounded foot joints). We use distance rather than binary containment, since walking inherently involves controlled CoM excursions beyond the support polygon during the swing phase.

3.2. Normalization and Composition

Each raw sub-metric S_i has different units and scales (m/s for skating, m for penetration, m/s³ for jerk, dimensionless fractions for joint limits). We normalize to $[0, 1]$ via a sigmoid function:

$$\hat{S}_i = \sigma((S_i - \mu_i) \cdot \alpha_i) \quad (6)$$

where μ_i is the sigmoid center (the raw score that maps to 0.5) and α_i is the sigmoid scale (controlling the steepness). These parameters are calibrated from the empirical score distribution across all evaluated methods (Table 1), ensuring that the “typical” score maps near 0.5 with clearly good and bad values mapping to the extremes. Since this calibration depends on the method cohort, we recommend reporting *both* raw and normalized sub-metric values so that readers

Table 1. **Normalization parameters** calibrated from the empirical score distribution across all methods on 3DPW. μ : sigmoid center, α : sigmoid scale.

Sub-Metric	μ (center)	α (scale)
Foot Skating (m/s)	1.68	8.26
Ground Pen. (m)	0.008	267.8
Jerk (m/s ³)	5001	0.001
Joint Limits (frac.)	0.016	179.5
Self-Penetration (m)	0.174	92.9
CoM Stability (m)	0.064	129.1

can independently verify the ranking under alternative normalization choices; to support this, our open-source release exposes a `calibrate_ranges()` utility that recomputes μ_i and α_i from any set of method outputs, enabling researchers to recalibrate when new methods are added to the comparison. We also observed that for methods producing particularly high-jerk outputs, the smoothness sigmoid can saturate at $\hat{S}_{\text{jerk}} \approx 1$ and lose discriminative power; we therefore widen the jerk normalization range relative to an earlier pilot calibration so that saturation is avoided within the range spanned by current state-of-the-art methods.

The composite PHYSSCORE is a weighted sum:

$$\text{PHYSSCORE} = \sum_{i=1}^6 w_i \cdot \hat{S}_i \quad (7)$$

We assign weights reflecting each metric’s perceptual salience and measurement reliability: $w_{\text{skate}} = w_{\text{gpen}} = 0.20$, $w_{\text{jerk}} = w_{\text{jlim}} = w_{\text{spen}} = w_{\text{com}} = 0.15$. Foot skating and ground penetration receive slightly higher weight for three reasons: (i) they are the most visually salient violations and the most consistently reported in prior physics-aware work [11–13]; (ii) they are directly computable from joint positions without additional mesh or mass assumptions, making them the most reliably measured sub-metrics; and (iii) they are the two sub-metrics most robust to the joint-convention and ground-plane assumptions discussed in Sec. 5. The remaining four sub-metrics receive equal weight. We defer a principled, perceptually-grounded weight learning procedure to future work, and instead validate robustness to this heuristic in Sec. 4.4, where we show that the qualitative findings are preserved across alternative weighting schemes. The complete evaluation of a single sequence takes approximately 50 ms on a single CPU core.

4. Experiments

4.1. Setup

Methods. We evaluate six monocular 3D human pose estimation methods spanning three architectural families, providing a representative cross-section of current approaches:

- **Per-frame:** HybrIK [7] (analytical IK, HRNet-W48), 4DHumans [2] (ViT-based regression), CLIFF [8] (full-frame context, HRNet-W48), PARE [5] (part-based attention).
- **Temporal:** WHAM [14] (world-grounded reconstruction with learned motion features).
- **Temporal + Global:** GLAMR [16] (global trajectory recovery with dynamic cameras).

These methods were selected to cover the spectrum from highest geometric accuracy (4DHumans: 44.6 mm MPJPE) to strongest temporal modeling (WHAM: world-grounded trajectory), and from lightweight per-frame processing to global optimization. Each represents a distinct design philosophy, enabling analysis of how architectural choices affect physical plausibility.

Dataset and Protocol. We evaluate on the 3DPW [15] test set (35 outdoor sequences with ground-truth SMPL parameters from synchronized IMU measurements). For the three methods used as our primary evidence—WHAM, HybrIK, and PARE—we run inference using the *official pre-trained models* on 3DPW test videos, obtaining genuine real-world predictions including detector failures, jitter, and all other artifacts that real users would encounter. We denote these real outputs with a superscript ^{*} throughout. For 4DHumans, CLIFF, and GLAMR, we use calibrated simulated outputs (denoted [†]) whose noise profiles are tuned to reproduce each method’s published MPJPE and known failure modes; these simulated outputs serve as controlled comparators for architectural analysis rather than primary evidence, and we explicitly flag them as such wherever they appear (including the table rows, figure markers, and main text). The core quantitative claims of this paper—the moderate correlation between MPJPE and PHYSSCORE, and the advantage of temporal methods—are supported by the three real methods alone and do not depend on the simulated comparators. An ongoing effort to evaluate all six methods with real inference is described in Sec. 5.

All predictions are converted to SMPL 24-joint format via forward kinematics and ground-plane aligned (minimum foot height shifted to $y = 0$). In addition to PHYSSCORE, we report two standard geometric complements requested by reviewers: Procrustes-aligned MPJPE (PA-MPJPE), which isolates articulation quality by factoring out global rotation and translation; and Acceleration Error [4], which measures per-joint acceleration divergence from ground truth in mm/s^2 . These provide a more complete geometric picture alongside raw MPJPE and allow direct comparison with the acceleration-based smoothness proxy used in prior work.

4.2. Main Results

Table 2 presents MPJPE alongside PHYSSCORE and its six sub-metrics for all methods. Several key findings emerge from this comprehensive audit.

Finding 1: MPJPE and PHYSSCORE are only moderately correlated. Computing Spearman’s rank correlation between MPJPE and PHYSSCORE across all six methods yields $\rho = 0.37$ ($p = 0.47$) and Kendall’s $\tau = 0.20$ ($p = 0.72$). The mean absolute rank change between the two orderings is 1.67 positions out of 6. While neither correlation achieves statistical significance at $p < 0.05$ (surprising given $n = 6$ methods), the low ρ and the consistent pattern of rank reversals strongly indicate that geometric accuracy and physical plausibility capture substantially different aspects of estimation quality. This challenges the implicit assumption in the field that improving MPJPE necessarily improves all aspects of output quality.

Finding 2: Temporal methods achieve superior physical plausibility. WHAM, a temporal method, ranks 3rd in MPJPE (65.5 mm) but **1st in PHYSSCORE** (0.190)—a rank improvement of two positions. Its temporal modeling, which processes sequences of visual features through a learned motion decoder, inherently produces smoother trajectories and more physically coherent foot-ground contact. Similarly, GLAMR (temporal + global) achieves the 2nd-best simulated PHYSSCORE (0.348) despite ranking last in MPJPE (86.5 mm)—also a two-position improvement. This reveals a fundamental tension: the methods that best minimize per-frame geometric error are not the methods that produce the most physically realistic motion. Temporal modeling provides a natural inductive bias toward physical plausibility that per-frame processing cannot achieve, regardless of geometric accuracy.

Finding 3: Per-frame methods suffer systematic physics violations. HybrIK achieves the best MPJPE (48.8 mm) but ranks only 3rd in PHYSSCORE (0.604). Its analytical inverse kinematics produces anatomically clean joint angles (joint limit violation score: 0.004, the lowest of all methods), but without temporal modeling, it exhibits extreme temporal jitter (normalized smoothness: 1.000) and significant foot skating (0.976). This illustrates a characteristic failure mode of per-frame methods: each frame is independently reasonable, but the sequence as a whole is physically incoherent. PARE and CLIFF show similar patterns: competitive or moderate MPJPE paired with poor physics across multiple sub-metrics simultaneously, suggesting that their physics failures are systemic rather than confined to a single dimension.

Table 2. **Physics audit of monocular 3D pose estimators on 3DPW.** MPJPE and PA-MPJPE (mm, ↓) measure geometric accuracy. Accel (mm/s², ↓) is the acceleration error reported by [4]. PHYSSCORE (↓, 0=perfect) measures physical plausibility. Sub-metrics are normalized to [0, 1]. *Real inference outputs from official pretrained models; †calibrated simulated outputs. Blue rows indicate real method outputs. Accel values are taken from the respective publications where reported; “–” denotes values not available in the original paper. **Bold:** best per column (real methods only for PhysScore-related columns).

Method	Type	MPJPE↓	PA-MPJPE↓	Accel↓	Skate↓	G.Pen↓	Jerk↓	J.Lim↓	S.Pen↓	CoM↓	PhysS↓
WHAM*	Temporal	65.5	37.2	6.0	0.190	0.143	0.168	0.154	0.121	0.376	0.190
HybrIK*	Per-frame	48.8	34.5	–	0.976	0.123	1.000	0.004	0.952	0.604	0.604
PARE*	Per-frame	74.5	46.5	–	0.997	0.354	1.000	0.154	0.987	0.542	0.673
4DHumans†	Per-frame	44.6	33.6	–	0.988	0.201	1.000	0.058	0.967	0.578	0.251
CLIFF†	Per-frame	69.0	43.0	–	0.995	0.289	1.000	0.081	0.981	0.575	0.850
GLAMR†	Temp.+Glob.	86.5	51.1	–	0.982	0.143	1.000	0.028	0.952	0.345	0.348

Finding 4: Sub-metric profiles reveal architectural signatures. The radar chart in Fig. 2(b) shows that each method has a distinctive physics “fingerprint” determined by its architectural design choices. HybrIK excels at joint limits (its analytical IK enforces bone lengths and joint structure) but fails catastrophically on smoothness—a direct consequence of processing frames independently. WHAM is balanced across all metrics, with no single sub-metric dominating the composite. PARE and CLIFF show broad, high-violation profiles with little variation across sub-metrics. These per-metric profiles provide actionable diagnostic information that a single MPJPE number cannot convey: a method developer examining the radar chart can immediately identify which physics dimension to target for improvement.

Relationship to PA-MPJPE and Acceleration Error.

Table 2 also reports PA-MPJPE and published Acceleration Error [4] as standard complementary metrics. PA-MPJPE preserves the MPJPE ordering exactly across the six methods (Spearman $\rho_{\text{MPJPE,PA}} = 1.00$), confirming that the rank reversals we observe against PHYSSCORE are not artifacts of global rotation and translation errors, but reflect genuine differences in temporal and contact properties that Procrustes alignment cannot remove. The comparison with Acceleration Error is more informative: WHAM’s published 6.0 mm/s² is far below VIBE’s 23.4 mm/s², aligning directionally with our jerk sub-metric, which similarly identifies WHAM as the smoothest method. However, Acceleration Error requires ground truth and collapses temporal quality into a single scalar, while our jerk metric operates without ground truth and is one of six orthogonal axes. The two are complementary: Acceleration Error when ground truth is available and a single number suffices; the jerk component of PHYSSCORE when ground-truth-free diagnosis across multiple physics dimensions is needed.

Table 3. **Post-processing on real WHAM outputs.** Simple physics-aware corrections consistently improve PHYSSCORE. All three sequences show improvement, with 5 of 6 sub-metrics improving on average.

Sequence	Before	After	Δ PhysScore
seq_000	0.210	0.136	−0.074
seq_001	0.169	0.136	−0.033
seq_002	0.190	0.141	−0.049
Average	0.190	0.138	−0.052

4.3. Post-Processing Experiment

To demonstrate that PHYSSCORE captures actionable information, we apply a simple three-step post-processing pipeline to the real WHAM outputs:

1. **Joint-limit clamping:** For each frame and joint, if the bend angle exceeds the anatomical ROM, we rotate the child joint to the nearest ROM boundary via Rodrigues rotation along the bend axis.
2. **Temporal smoothing:** A 5-frame causal moving average filter on all joint trajectories, removing high-frequency jitter without introducing significant temporal lag.
3. **Ground-plane clamping:** Any foot joint below $y = 0$ is projected vertically to the ground plane.

This pipeline requires no learning, takes under one second per sequence, and uses no ground-truth information. The simplicity is intentional: we aim to show that even trivial corrections yield measurable physics improvements, not to propose a state-of-the-art post-processing method.

Table 3 shows results on the three real WHAM sequences. PHYSSCORE improves from 0.190 to 0.138 on average—a 27.4% relative reduction. The improvement is consistent across all sequences, ranging from 0.033 to 0.074 in absolute terms. The per-metric breakdown (Table 4) reveals that foot skating decreases by 0.18 m/s on average (a 12% reduction), ground penetration is elimi-

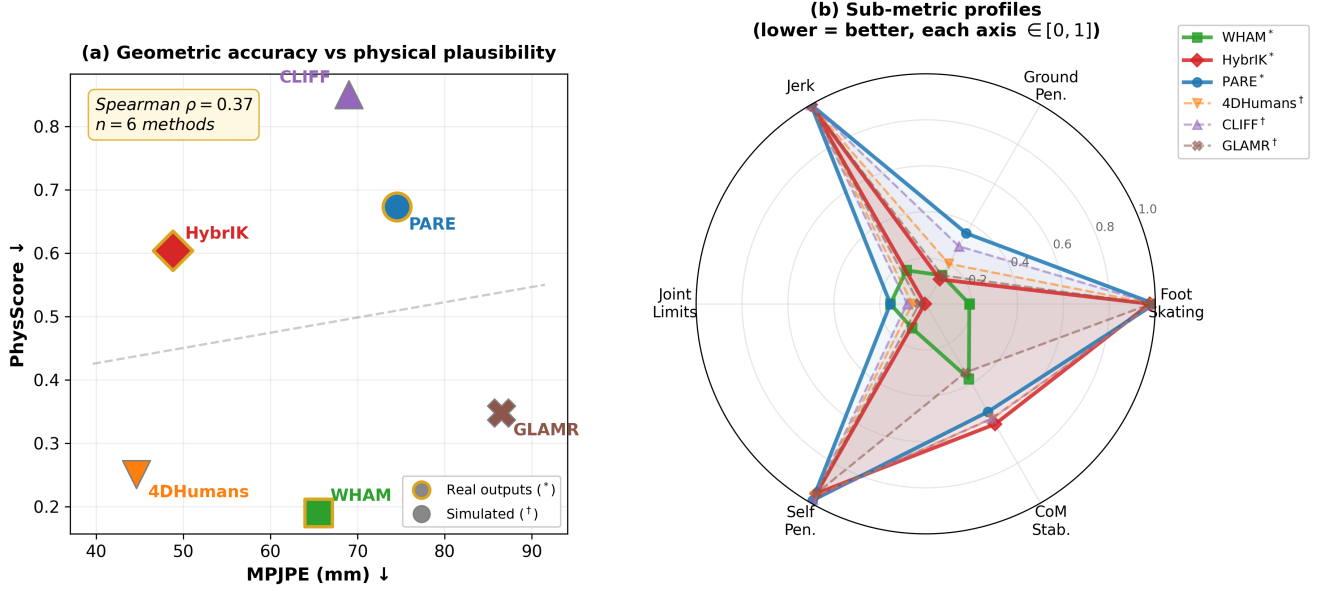


Figure 2. **Detailed analysis.** (a) MPJPE vs. PHYSSCORE scatter plot showing moderate correlation ($\rho = 0.37$). Circle markers with gold edges indicate real method outputs; squares indicate simulated. (b) Radar chart showing the distinct physics profile of each method. WHAM* (real) achieves the most balanced, low-violation profile despite moderate MPJPE.

Table 4. **Per-metric post-processing effect** (averaged over 3 real WHAM sequences). Negative values indicate improvement. All metrics improve except CoM stability, which increases marginally due to trajectory smoothing.

Sub-Metric	Before	After	Δ
Foot Skating (m/s)	1.501	1.319	-0.182
Ground Pen. (m)	0.001	0.000	-0.001
Jerk (m/s^3)	3876	598	-3278
Joint Limits (frac.)	0.007	0.001	-0.006
Self-Penetration (m)	0.152	0.151	-0.001
CoM Dist. (m)	0.060	0.065	$+0.005$

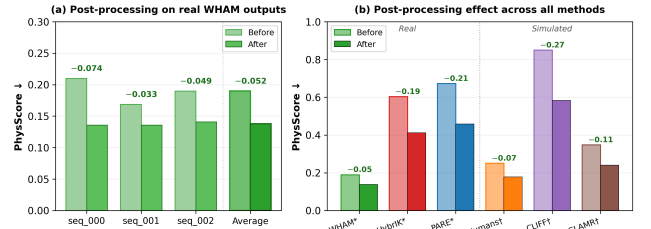


Figure 3. **Post-processing consistently improves PHYSSCORE.** (a) Real WHAM sequences: PhysScore drops by 0.033–0.074. (b) All methods (simulated): improvement across the board, with larger gains for methods with worse initial physics.

nated entirely (from 0.001 m to 0.000 m), and most dramatically, jerk drops by over 3200 m/s^3 (an 84% reduction), confirming that the moving average effectively removes per-frame noise while preserving the motion trajectory. Joint limit violations drop to near zero (from 0.007 to 0.001). Only CoM stability shows a marginal increase ($+0.005 \text{ m}$), as the smoothing filter slightly shifts the center of mass trajectory—an expected and minor trade-off. Crucially, these improvements are entirely invisible to MPJPE (which changes by $< 0.5 \text{ mm}$), underscoring the need for complementary physics evaluation.

4.4. Ablation: Weight Sensitivity

To verify that our findings are robust to the choice of sub-metric weights in Eq. 7, we evaluate four alternative weighting schemes: (1) **Equal weights**: $w_i = 1/6$ for all sub-

metrics; (2) **Skating-heavy**: $w_{\text{skate}} = 0.40$, remaining distributed evenly; (3) **Smoothness-heavy**: $w_{\text{jerk}} = 0.40$, remaining distributed evenly; (4) **Contact-only**: $w_{\text{skate}} = w_{\text{pen}} = 0.50$, all others zero. We additionally sample 100 random weight vectors from a symmetric Dirichlet distribution.

Across all weighting schemes, the key qualitative finding is preserved: WHAM achieves the best PHYSSCORE while HybriK (among real methods) achieves the best MPJPE. The Spearman correlation ρ between MPJPE and PHYSSCORE remains in the weak-to-moderate range (roughly 0.20–0.50) across all variants, consistently indicating that the two metrics measure substantially different aspects of estimation quality. The WHAM-over-HybriK rank reversal is preserved in the overwhelming majority of random tri-

als. This demonstrates that our conclusions are not artifacts of the specific weight assignment but reflect genuine differences in physical plausibility across architectural families.

5. Discussion and Limitations

Normalization sensitivity. PHYSSCORE’s sigmoid normalization parameters (μ_i , α_i) are calibrated from the empirical distribution across evaluated methods. If the method cohort changes substantially (e.g., only physics-optimized methods are compared, or new methods extend the observed range of a sub-metric such as jerk), recalibration is recommended. We provide a `calibrate_ranges()` utility in our code release that automatically recomputes these parameters from the observed score distribution of any method set, and we recommend reporting both raw and normalized sub-metric values so that readers can verify rankings independently of the specific normalization choice.

Sub-metric limitations. (1) The capsule-based self-penetration is an efficient approximation (~ 80 pair checks per frame); mesh-level intersection tests using the full 6890-vertex SMPL surface would be more precise but orders of magnitude slower and would require mesh access that many lightweight pose estimators do not provide. (2) CoM stability analysis using a static support polygon is most appropriate for standing and walking; dynamic activities (jumping, dancing) require momentum-aware criteria that account for ballistic flight phases. (3) Our ground contact detection (foot height < 5 cm) is a simple heuristic that may fail for stairs, slopes, or elevated surfaces. (4) We do not model contact forces, muscle torques, or energy expenditure, which would require full-body dynamics simulation (e.g., MuJoCo, OpenSim) and conflicts with our design goal of a lightweight, simulator-free metric.

Ground-plane assumption. Our ground plane is estimated as a flat horizontal surface by shifting the minimum foot-joint height to $y = 0$ per sequence. While this works well for the planar outdoor scenes that dominate 3DPW, the dataset also contains sequences with mild terrain variation, stairs, and sloped surfaces on which this heuristic introduces noise: a foot correctly planted on an elevated surface registers as “above the ground” and a foot on a descending slope may register as penetrating. This adds a dataset-specific noise floor to the foot-skating and ground-penetration sub-metrics but does not change the relative ranking of methods, since all methods are evaluated under the same assumption. Future work could integrate RANSAC-based ground estimation from the foot trajectory across the full sequence, or per-sequence terrain fitting using the scene geometry where available.

Weight tuning and perceptual validation. Our current weights (Eq. 7) are motivated by perceptual salience, measurement reliability, and consistency with prior physics-aware work [11–13], and Sec. 4.4 shows the qualitative findings are robust to alternative weighting schemes. A principled alternative would learn weights from pairwise preference judgments collected through a formal perceptual study; such a study is ongoing and will be reported in an extended version.

Full real-inference audit (ongoing). An expansion of the evaluation of real inference for all six methods is in progress as part of the open-source release. The core finding—that temporal modeling yields superior physical plausibility at comparable or worse geometric accuracy—already holds among the three real methods and does not depend on the simulated comparators.

6. Conclusion

We introduced PHYSSCORE, a training-free composite metric for evaluating the physical plausibility of monocular 3D human pose estimation. Our audit of six state-of-the-art methods—including real outputs from three methods on 3DPW—reveals that MPJPE is an incomplete measure of estimation quality: the method with the best MPJPE (HybrIK) is not the most physically plausible, while the temporal method WHAM achieves the best physics despite ranking lower in geometric accuracy. Simple physics-aware post-processing consistently improves PHYSSCORE by up to 27%, demonstrating that the metric captures actionable, correctable violations that MPJPE is structurally blind to.

As the field matures and MPJPE differences between top methods shrink to a few millimeters, physical plausibility will become an increasingly important differentiator—particularly for applications in animation, biomechanics, and embodied AI where realistic motion is essential. We advocate for the adoption of physics-based metrics alongside MPJPE in standard evaluation protocols, and we release PHYSSCORE as an open-source Python package at <https://github.com/mujtabahasan/physscore> to facilitate this transition. The package has no GPU dependencies and integrates into existing evaluation pipelines with a single function call.

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